

Computer Networks

Lecture 18 Data Link Layer

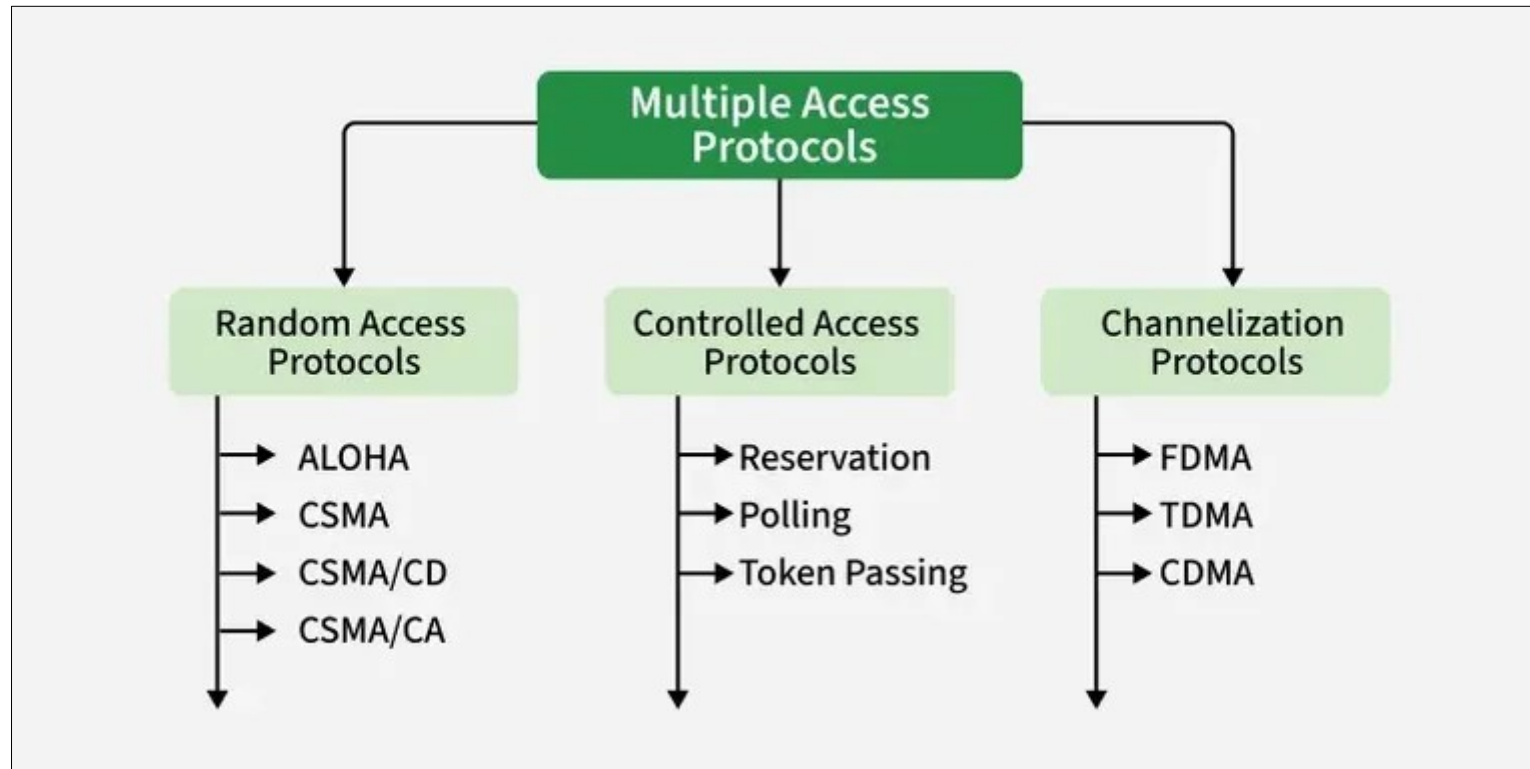
Content



- **Controlled Access Protocol**

Introduction

Many formal protocols have been devised to handle access to a shared link



Scheduling

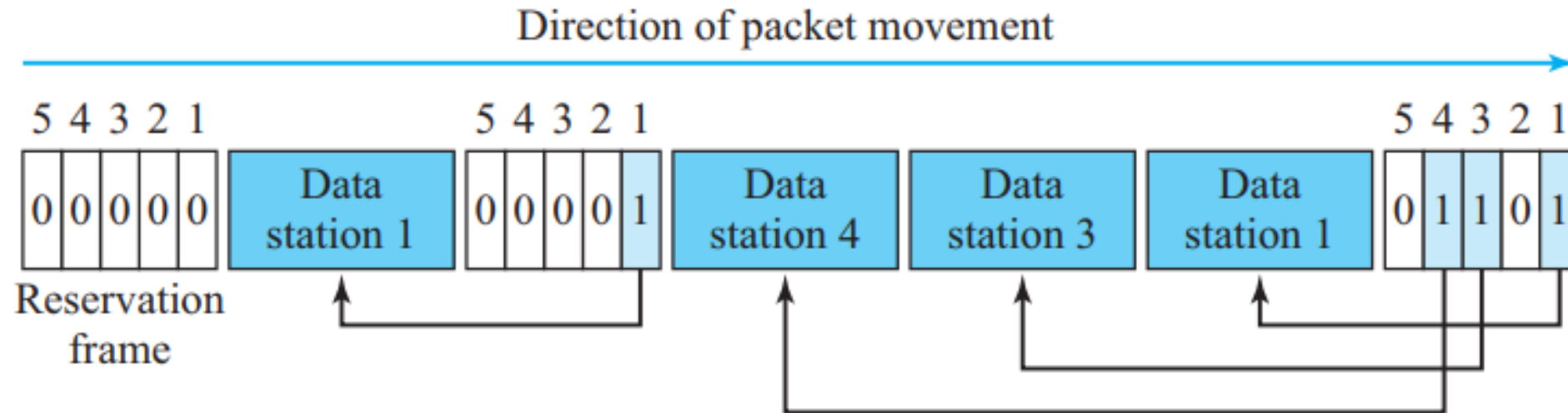
Schedule frame transmissions to avoid collision in shared medium

- ✓ More efficient channel utilization
- ✓ Less variability in delays
- ✓ Can provide fairness to stations
- ✗ Increased computational or procedural complexity

- Three major approaches
 - Reservation
 - Polling
 - Token Passing

Reservation

In the reservation method, a station needs to make a reservation before sending data. Time is divided into intervals. In each interval, a reservation frame precedes the data frames sent in that interval.



- Transmissions organized into cycles
- Cycle: reservation interval + frame transmissions

Reservation System Options

Centralized or distributed system

- *Centralized systems*: A central controller listens to reservation information, decides order of transmission, issues grants
- *Distributed systems*: Each station determines its slot for transmission from the reservation information

Single or Multiple Frames

- *Single frame reservation*: Only one frame transmission can be reserved within a reservation cycle
- *Multiple frame reservation*: More than one frame transmission can be reserved within a frame

Channelized or Random Access Reservations

- *Channelized reservation*: Reservation messages from different stations are multiplexed without any risk of collision
- *Random access reservation*: Each station transmits its reservation message randomly until the message goes through

Channelized Reservation

In Channelized Reservation, the reservation phase is divided into fixed slots. Each station is assigned a dedicated reservation mini-slot.

Steps

- The reservation frame contains one mini-slot per station.
- Each station sends a 1-bit reservation request in its own slot.
- Since each station has a separate slot, collisions cannot occur.
- Stations that send reservation requests are allowed to transmit in the data transmission phase.

Example

Reservation Phase

Slot	Station	Reservation Bit
1	S1	1
2	S2	0
3	S3	1
4	S4	0

S1 and S3 want to transmit.

Data Transmission Phase:

Data Slot	Station
1	S1
2	S3

Efficiency of Reservation Systems

Single Frame reservation Scheme

- Each data frame transmission time = X seconds
- Each reservation minislot duration = vX seconds
- There is one reservation frame followed by one data frame in each cycle.

$$\begin{aligned} \text{Total time} &= X + vX \\ &= (1 + v)X \end{aligned}$$

$$\text{Efficiency} = \frac{\text{Useful data transmission time}}{\text{Total cycle time}} \Rightarrow \frac{X}{(1+V)X} \Rightarrow \frac{1}{1+V}$$

Efficiency of Reservation Systems

K – Frame reservation Scheme

- Each data frame transmission time = X seconds
- Each reservation minislot duration = vX seconds
- The reservation phase is followed by K data frames..

$$\begin{aligned} \text{Total time} &= KX + vX \\ &= (K + v)X \end{aligned}$$

$$\text{Efficiency} = \frac{\text{Useful data transmission time}}{\text{Total cycle time}} \Rightarrow \frac{KX}{(K+V)X} \Rightarrow \frac{K}{K+V}$$

Slotted Aloha Reservation Scheme

Efficiency The maximum throughput of Slotted ALOHA occurs when the offered load $G = 1$.

Throughput:

$$S = Ge^{-G}$$

At $G = 1$:

$$S = \frac{1}{e}$$

- Only $1/e$ of the attempts succeed.
- Therefore, on average $e = 2.71$ attempts are required for one successful reservation.

Thus the effective reservation time:

$$\begin{aligned}\text{Reservation time} &= e \times vX \\ &= 2.71vX\end{aligned}$$

$$\text{Efficiency} = \Rightarrow \frac{1}{1+2.71V}$$

Polling

Polling works with topologies in which one device is designated as a primary station and the other devices are secondary stations.

Instead of stations competing for the channel (as in random access methods), the primary station sequentially asks each secondary station whether it wants to transmit data.

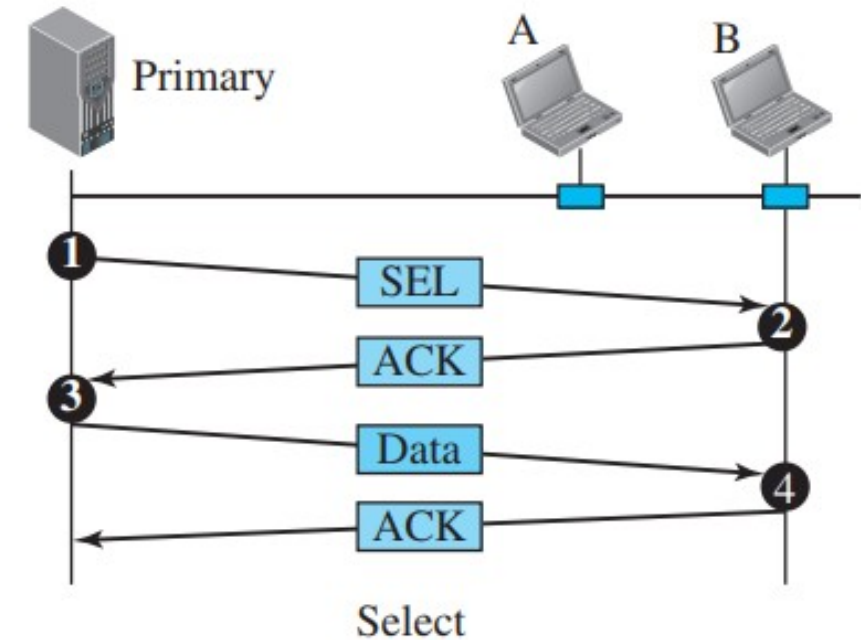
Steps

- A primary station controls the communication channel.
- The primary station sends a poll message to a secondary station.
 - The secondary station replies:
 - If it has data → it transmits the data.
 - If it has no data → it sends a “no data” response or remains idle.
- The primary station then polls the next station in sequence.

Polling

Select Function

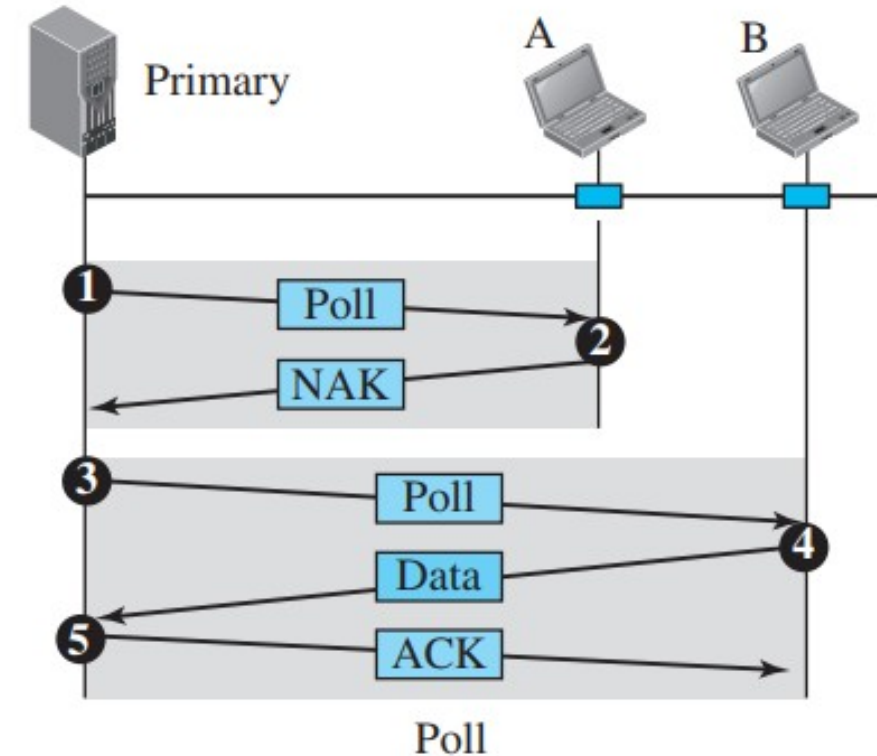
- The select function is used when the primary device wants to send data to a secondary device.
- The primary first sends a Select (SEL) frame containing the address of the intended secondary.
- This alerts the secondary and checks whether it is ready to receive data.
- After receiving an acknowledgment from the secondary, the primary transmits the data.



Polling

Poll Function

- The poll function is used by the primary device to check if secondary devices have data to send.
- The primary polls each secondary device one by one.
- The secondary responds with NAK if it has no data or a data frame if it has data.
- If data is received, the primary sends an ACK to confirm successful reception.



Polling

Service Limit How much is a station allowed to transmit per poll?

- Exhaustive: until station's data buffer is empty (including new frame arrivals)
- Gated: all data in buffer when poll arrives
- Frame-Limited: one frame per poll
- Time-Limited: up to some maximum time

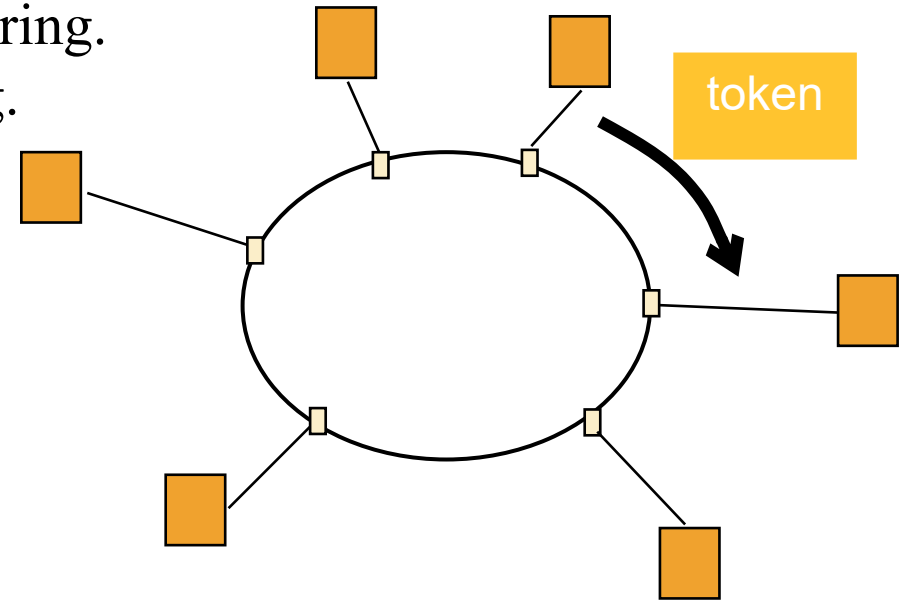
Priority: Priority can be given to some stations so that important stations get faster access to the channel.

- Stations that require higher priority are placed multiple times in the polling sequence. This gives them more chances to transmit, resulting in more bandwidth and lower delay.

Token Passing

In the token-passing method, the stations in a network are organized in a logical ring.

- Predecessor: The station before the current station in the ring.
- Successor: The station after the current station in the ring.
- Current station: The station currently using the channel.

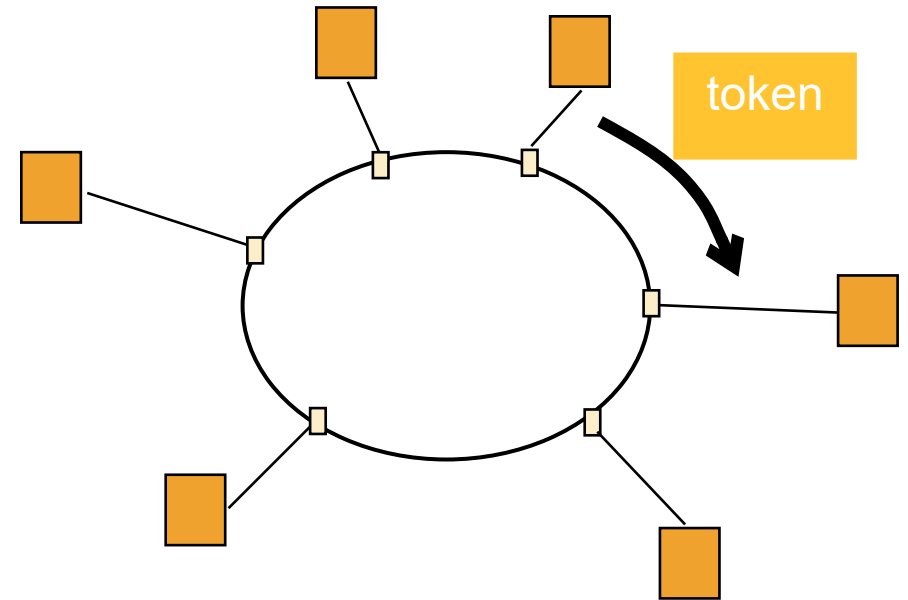


The right to access the channel is passed from the predecessor to the current station. After finishing transmission, the current station passes the right to the successor.

Token Passing

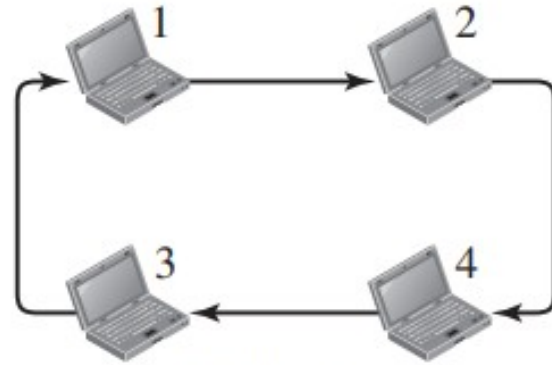
Steps

- A token (special control frame) continuously circulates around the ring.
- A station waits until it receives the token.
- If the station has data, it captures the token and sends the data frame.
- The data frame travels around the ring to the destination station.
- After transmission, the token is released back into the ring for the next station.

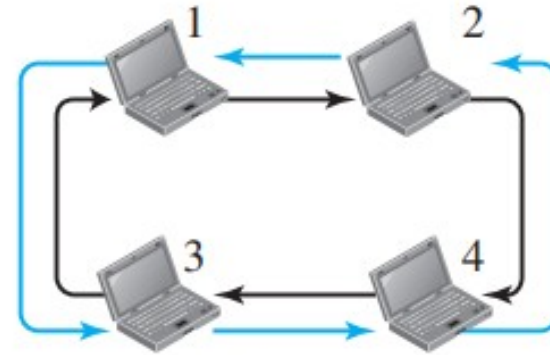


Token Passing

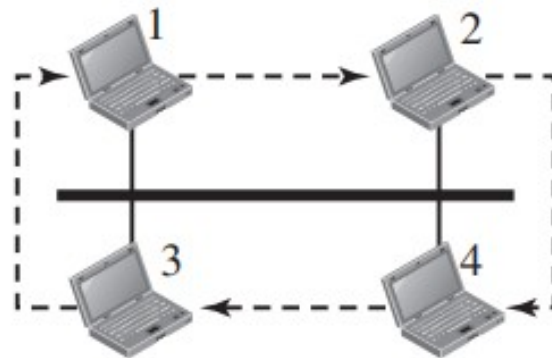
Ring Types



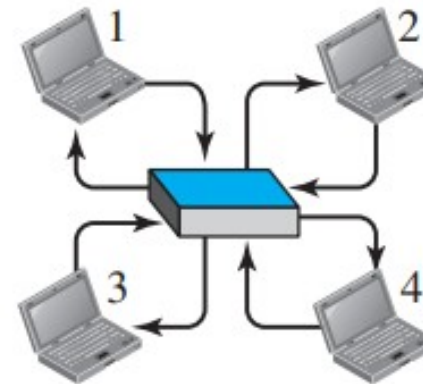
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

Question

Q 1. A reservation system uses minislots of duration 0.1 ms and a data frame transmission time of 2 ms. Calculate the maximum efficiency

Q 2. In a K-frame reservation scheme, minislot duration = 0.05 ms, data frame time = 1 ms, and $K = 4$. Calculate efficiency.

Q 3. Find the effective reservation time in a Slotted ALOHA reservation scheme where minislot duration = 0.2 ms.